

Recommendation Supervision Semantics and Exposure-Aware Evaluation for Recommendation-Tuned LLMs

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Abstract—Empirical conclusions about large language models (LLMs) for recommendation can depend on what is supervised, how much downstream supervision a model consumes, and how candidates are evaluated. We study these dependencies using Llama-3.2-3B-Instruct with QLoRA, taking MovieLens-1M as the main dataset and Amazon Musical Instruments as a directional ranking-side check. We compare rating-derived preference prediction (Y), next-interaction prediction (N), and a shared Y+N multitask adapter (M1), trace task-sample exposure, evaluate alternative candidate protocols, and compare N-trained LlamaRec with SASRec at approximately matched downstream exposures. The complete Y and N formulations yield distinct measured capability profiles. In the MovieLens seed42 trajectory, Y’s measured gains through 96k are limited or uneven, whereas N ranking improves across measured validation and test points through 200k. On a common cross-task-safe subset, the standard-k5 N-specialist-M1 validation gap narrows from 48k to 96k per task across all three metrics, although test does not reproduce the narrowing. At 96k, three training seeds retain a small but consistently positive N advantage, while M1 preserves comparable Y-side preference capability. Alternative candidate protocols materially change the remaining specialist-multitask gap. N-trained LlamaRec leads the evaluated SASRec baseline at four approximately matched downstream task-sample exposure points, but SASRec improves strongly and the gap narrows at 200k. Safe ranking directions also appear on Amazon at earlier seed42 operating points. These results show that recommendation capability and relative model performance are conditional on prediction formulation, cumulative downstream exposure, and candidate evaluation protocol.

Index Terms—Large language model recommendation, Recommendation supervision, Sequential recommendation, Multitask learning, Task-sample exposure

I. INTRODUCTION

Language-model-based recommendation spans zero-shot conversation, candidate ranking, and recommendation-specific adaptation [1]–[3]. Recommendation-as-language-processing further offers a shared language interface for model inputs and predictions [4]. Because language models can interact, rank, or adapt to a task, interpreting what recommendation-specific supervision teaches them is a central empirical problem [5].

Recommendation includes different prediction questions. Whether a user likes an item is a preference judgment; which item appears next is a subsequent-event prediction. Feedback and sequential models operationalize different targets, while language frameworks can express both through a common interface [4], [6], [7]. For an adapted model, task data, prompt, target, and scoring interface jointly define the complete supervision formulation and the capability observed.

A checkpoint also represents a specific amount of adaptation. We define task-sample exposure as the cumulative number of examples consumed for a task, including repetitions. Unified recommendation studies already examine task performance and changes in data amount or composition [4], [8]. Our complementary question is how specialist-shared relationships evolve as the corresponding task receives more cumulative supervision, with total and per-task exposure separated for mixed-task training.

Evaluation conditions add another boundary. Sampled metrics, sampling strategies, and candidate construction can change measured performance and model orderings [9]–[11]. Task formulation and data allocation likewise intersect in unified recommendation and shared search-recommendation models [4], [8], [12], [13]. These precedents motivate jointly examining cumulative per-task supervision, specialist-shared

modeling, and candidate protocol without treating any one as an isolated cause.

We conduct a controlled study with rating-derived preference supervision (Y), next-interaction supervision (N), and one shared Y+N adapter (M1). We trace task-specific exposure, compare specialists with M1 at matched per-task exposure, test alternative candidate protocols, and compare N-trained LlamaRec with SASRec at approximately matched downstream supervision. MovieLens supplies detailed trajectories; Amazon Musical Instruments provides an external ranking-side check.

The complete Y and N formulations produce distinct measured capability profiles. In the seed42 MovieLens trajectory, Y gains are limited or uneven, whereas N improves through the evaluated range. On a common cross-task-safe subset, the standard-k5 validation specialist-M1 gap narrows from 48k to 96k per task across all three metrics; two additional seeds reproduce a small positive N advantage at 96k, but the common seed42 test subset does not reproduce the narrowing. Alternative protocols retain an N advantage with protocol-dependent magnitude. N-trained LlamaRec leads SASRec at four approximately matched exposure points, while SASRec improves and the gap narrows at 200k. Amazon supports safe earlier-point ranking directions only.

Our primary contribution is an exposure-conditioned account of specialist-shared modeling. The secondary contribution is a four-point cumulative downstream-exposure comparison between the LLM and a sequential baseline. Candidate-protocol analysis supplies a supporting boundary, while supervision-formulation distinctions frame the capabilities being compared. Together, they show that conclusions depend on what is supervised, how much task supervision is consumed, and how candidates are evaluated.

II. RELATED WORK

A. LLMs for Recommendation

Prompt-based recommendation places language models directly in recommendation requests. Zero-shot conversational recommendation and zero-shot ranking emphasize different outputs: interaction and item ordering [1], [2]. Within the broader LLM-recommender literature [5], this diversity makes the requested output part of what an evaluation measures and motivates task-specific capability claims.

Recommendation-specific adaptation makes the training interface part of the model’s role. P5 expresses multiple task families as language pairs, TALLRec trains Yes/No recommendation responses, and InstructRec supports pointwise, pairwise, matching, and reranking requests [3], [4], [14]. These precedents establish language-based adaptation across tasks. We compare the capabilities measured after complete preference and next-interaction formulations under a shared model setting.

B. Recommendation Supervision and Task Formulation

Recommendation targets differ before any language interface. Matrix factorization fits ratings, implicit-feedback methods distinguish preference from confidence in observations, and sequential models predict a subsequent item [6], [7], [15]. Their scores therefore concern assessments, observed behavior, or next-event selection. Separating target from modeling technique clarifies which capability a score supports.

A common language interface can retain these distinctions. P5 includes rating-threshold like/dislike questions and next-item interfaces; ITDR also combines rating and next-item tasks [4], [8]. Task distinction is therefore not new. Our framing compares their observed native and bridge capability profiles while treating target, task data, prompt, and scoring rule as one complete formulation rather than a single isolated semantic factor.

C. Multitask and Unified Recommendation Modeling

Unified recommendation already links task design, shared modeling, and data allocation. P5 varies derived instances while comparing single- and multitask models; ITDR reports task-removal and data-subset studies; OpenOneRec specifies task-sampling weights and contrasts low-data, single-domain, and multidomain adaptation [4], [8], [12]. These are substantive data-aware multi-task precedents. Our narrower question tracks specialist-shared relationships as corresponding-task cumulative exposure changes.

Specialization also depends on the combined tasks or domains. Penha et al. vary added-task per-item caps when comparing specialized and shared search-recommendation models, while OneReason studies reinforcement-learning experts and a unified student [13], [16]. Rather than claiming the first data-aware specialist-shared comparison, we align cumulative exposure for each Y/N target task and then test how the resulting ranking relationship varies across candidate protocols.

D. Evaluation and Baseline Comparison

Candidate construction can alter offline ranking conclusions: sampled metrics need not preserve exact-metric orderings, and sampling strategy can affect measurements and model rankings [9], [11]. Target-set construction and difficult retrieved candidates provide related precedents [10], [14]. Candidate sensitivity is therefore established. We use separate non-nested protocols to delimit the higher-exposure specialist-multitask relationship, without isolating candidate count from composition or sampling.

Training conditions complement evaluation scope. SASRec is a sequential reference, while replicability and standardized-evaluation studies underscore careful tuning and the competitiveness of conventional baselines [7], [17], [18]. TALLRec already compares an adapted LLM with SASRec under selected-example budgets, while sequencescaling work studies data-pool size and repetition [3], [19]. Our contribution is not the first budget-aware comparison: it follows both models over four approximately matched cumulative downstream-consumption points, a quantity distinct from selected examples and computation.

Prior work therefore establishes task diversity, shared modeling, data-dependent performance, and candidate-sensitive evaluation. We connect these themes by tracing specialist-shared relationships over cumulative per-task supervision, using a sequential-baseline trajectory and alternative candidate protocols as complementary comparisons and boundaries. Supervision formulation specifies the prediction capability to which each conclusion applies.

III. PROBLEM FORMULATION

Let an interaction be a user-item event with a rating and timestamp, and let $H(u,t)$ contain a user’s interactions strictly before t . Samples use the complete sequence, while model input retains at most the 10 most recent strictly earlier interactions, $H_{10}(u,t)$, paired with an item i or candidate set C . A rating-derived preference label and the next observed item answer different questions even when drawn from the same sequence.

A. Preference, Next Interaction, and Multitask Interfaces

Y estimates $P(\text{Like} \mid H_{10}, i)$, where ratings at least 4 map to Yes and lower ratings to No. The complete sequence, including low-rated events, supplies histories and targets. Native evaluation measures rating-derived class discrimination; independently scoring candidates through $P(\text{Yes})$ provides the bridge to next-interaction ranking.

N estimates $P(\text{NextInteraction} = i \mid H_{10}, C)$, where C contains the observed next item and fixed distractors; the answer is the target’s candidate label. Targets use all rating levels. A distractor is an alternative to the observed next event, not a confirmed dislike, and previously encountered items remain eligible.

M1 interleaves Y and N within one shared adapter while retaining task-specific prompts and answer spaces. M-Y and M-N denote its two scoring interfaces. Each is compared with the corresponding specialist at matched task exposure; M1 total exposure is the sum across tasks.

RQ1 compares the observed Y and N capability profiles; RQ2 traces their response to task-sample exposure. RQ3 asks how closely M1 approaches both specialists at matched per-task exposure, and RQ4 tests how the standard PopMatch-k5 relationship changes under alternative candidate protocols. RQ5 follows the exposure-conditioned comparison between N-trained LlamaRec and SASRec.

IV. METHODOLOGY

The pipeline constructs temporally legal examples, adapts a common base model with task-specific prompts, and scores allowed responses. Complete user sequences define examples, bounded recent histories form prompts, and the Y and N interfaces produce preference scores or candidate orderings.

A. Temporal Sample Construction

Each user’s sequence is grouped into ordered timestamp buckets. A target at t uses only events strictly before t , so same-bucket Y targets share the preceding history. The last bucket supplies Y test targets, the previous bucket supplies validation, and earlier buckets supply training; a held-out bucket may yield several Y targets. Prompts retain the 10 most recent eligible events.

N requires preceding history and one unique next item, so legal targets are single-interaction buckets after the initial bucket. Multi-interaction buckets are skipped as targets but remain in later histories. The last two legal N examples become validation and test; earlier legal examples form training. Inputs again retain at most 10 strictly earlier interactions.

B. Shared Adaptation and Response Scoring

We adapt Llama-3.2-3B-Instruct [20] with QLoRA over a frozen 4-bit NF4 base [21]. Rank 16 LoRA adapters [22] use alpha 32 and dropout 0.05 across q/k/v/o and gate/up/down projections. Gradient checkpointing is enabled and maximum sequence length is 2,048.

Examples use instruction-chat format with the answer as the assistant response. Prompt and padding tokens are masked, giving response-only loss. Y supervises Yes/No and N supervises the target candidate label. M1 shares adapter parameters and interleaves Y/N examples 1:1, with the prompt identifying the task.

Inference scores allowed-answer likelihoods. Y normalizes Yes/No likelihoods and Y-as-ranker orders independently scored candidates. N normalizes candidate-label likelihoods conditioned on the list. Multi-token answers use complete-sequence log-likelihood; M-Y and M-N reuse these routes.

One-device micro-batch 1 with 8 accumulation steps yields 8 examples per optimizer update. Checkpoints are indexed by cumulative steps and task exposure and evaluated on fixed candidates. Validation guides training decisions; test follows frozen selection. Multitask per-task allocation is specified below.

V. EXPERIMENTAL SETUP

MovieLens-1M supports the exposure and specialist-multitask analyses; Amazon Musical Instruments supplies an external ranking check. Native preference and ranking metrics are reported separately with split and exposure identified. Validation guides exposure and checkpoint decisions, while frozen-test results are reported only after those decisions.

TABLE I Task-specific dataset splits.

Dataset	Task	Users	Train	Validation	Test
MovieLens-1M	Y	6040	976284	12381	11544
MovieLens-1M	N	5675	212725	5675	5675
Amazon Musical Instruments	Y	57439	396908	57442	57442
Amazon Musical Instruments	N	57439	339449	57439	57439

A. Datasets and Evidence Coverage

MovieLens-1M [23] uses task-specific temporal splits summarized in Table I. Different target and timestampbucket rules yield 6,040 Y users and 5,675 legal N users. Ratings cover 3,706 distinct items, while the metadata universe contains 3,883, a distinction retained for coverage interpretation.

The processed Amazon Reviews 2023 5-core Musical Instruments data [24] contain 57,439 users, 24,584 items, and 511,792 interactions; task splits appear in the same table. External evidence uses seed42 full-test Random-k5 and PopMatch-k5 ranking at available earlier operating points, covering Y-as-ranker, N, and sequential references.

B. Task-Sample Exposure and Matching

Task-sample exposure counts examples consumed for task q , including repetitions: $E_q = \sum_s \sum_{x \in B_s} 1[\text{task}(x)=q]$, where B_s is optimizer step s ’s batch. It measures received supervision, not unique interactions, because examples may repeat and task histories overlap.

With effective batch 8, specialist updates map directly to task exposure. Balanced M1-48 and M1-96 receive expected per-task exposures of 48k and 96k and totals of 96k and 192k; this accounting assumes correct resume data skipping. Table II separates optimizer steps, total exposure, and Y/N exposure.

TABLE II LLM optimizer steps and task-sample exposure (effective batch 8). For a specialist, total exposure equals its task exposure. M1 total exposure is twice its expected per-task exposure, conditional on correct resume data skipping.

Run	Steps	Total	Y exposure	N exposure
Y24	3000	24000	24000	0

Y48	6000	48000	48000	0
Y96	12000	96000	96000	0
N24	3000	24000	0	24000
N48	6000	48000	0	48000
N96	12000	96000	0	96000
N200	25000	200000	0	200000
M1-48	12000	96000	48000	48000
M1-96	24000	192000	96000	96000

N200 consumes 200,000 examples, about 94.0% of the 212,725-example legal N training pool, and serves as a near-one-pass anchor. SASRec uses recorded actual consumption, including short final batches: 24,064, 48,128, 96,256, and 200,000 examples. These points approximately match N24/N48/N96/N200 rather than equate computation.

C. Native Metrics and Candidate Protocols

Y-native evaluation reports AUC, F1, and accuracy; paired 96k F1/accuracy use threshold 0.5. Y-as-ranker independently scores candidates by P(Yes), whereas Nnative and M-N use candidate-label likelihoods. Ranking reports HR@1, NDCG@5, and MRR; NDCG is truncated at 5 and MRR uses the complete candidate ranking.

Random-k5 has one target and four random distractors. PopMatch-k5 draws distractors near the target’s full-corpus popularity from a pool up to 10 times the needed size, excluding the target. Fixed ordered lists are shared within comparisons. This retrospective control reduces one popularity mismatch; it is not online retrieval.

Hard-k20 and hard-k50 evaluate N96 and M1-96 on shared examples and ordered candidates within each protocol. Their

candidate sets are separately constructed and non-nested, so count and composition change together. They test protocol-conditioned robustness rather than isolate candidate size.

To ensure cross-task holdout integrity for the shared adapter, M1-N comparisons are restricted to a cross-tasksafe subset of N evaluation examples. For an example $e=(u,t)$, we retain it only when its target was not consumed as an M1 Y-training target, did not enter the history of any later M1 Y-training example, and no M1 Y-training target for user u has a timestamp at or after t . The M1 N-training branch has 0 target or history overlap with the retained N holdouts. N and M1-N are evaluated on exactly the same retained examples. Training follows the task-specific streams; this additional criterion is an evaluation-side restriction, not a joint temporal cutoff used during training.

Cross-task-safe coverage is 5,494/5,675 (96.81%) on M1-48 validation, 5,600/5,675 (98.68%) on its test, 5,318/5,675 (93.71%) on M1-96 validation, and 5,535/5,675 (97.53%) on its test Table III. For the seed42 comparison from 48k to 96k, we use the intersection of the M1-48 and M1-96 masks, giving the same 5,318 validation and 5,535 test examples at both exposure points.

TABLE III Cross-task-safe subset coverage for M1-N comparisons on MovieLens. The restriction is applied to evaluation, and N and M1-N use the same retained examples within each row.

M1 point	Split	Retained	Full holdout	Coverage (%)
M1-48	validation	5494	5675	96.81
M1-48	test	5600	5675	98.68
M1-96	validation	5318	5675	93.71
M1-96	test	5535	5675	97.53

D. Sequential Baseline and Comparison Scope

SASRec [7] uses 64-dimensional item and position embeddings, 2 attention heads, 2 causal Transformer layers, a 256-dimensional feed-forward block with GELU, dropout 0.2, final LayerNorm, and a maximum history of 10. It scores an item by the dot product between the sequence representation and item embedding plus an item bias. Training uses full-item cross-entropy and AdamW with learning rate 0.001, weight decay 0, no scheduler, batch size 512, and seed42.

S47, S94, S188, and S391 are predefined 47-, 94-, 188, and 391-optimizer-step operating points, independently trained from scratch rather than selected as best validation checkpoints or by early stopping. Their recorded downstream task-sample exposures are 24,064, 48,128, 96,256, and 200,000, including the short final batch. They are paired with N24/N48/N96/N200 on the same seed42 PopMatch-k5 validation and test candidate sets. Validation is the main comparison and frozen test is report-only. This design aligns downstream task-sample exposure only, not pretraining,

tokens, optimizer steps, FLOPs, wall-clock time, hardware, latency, parameter count, or computational resources.

E. Statistical Analysis and Training Seeds

Paired user-level bootstrap intervals [25] describe evaluation-sample uncertainty for the fixed seed42 Y96 and M1-Y96 models. Predictions share targets; users are resampled with replacement and all their examples enter both models’ resamples. Each resample yields the binary M1-Y minus Y96 metric difference, whose 2.5th and 97.5th percentiles form the 95% interval. These intervals support only the Y-side binary comparison; cross-task-safe M1-N ranking results are reported without bootstrap intervals.

Zero-crossing Y-side intervals indicate uncertainty, not equivalence. Full trajectories use training seed42; seeds 43/44 independently replicate Y96, N96, and M1-96 at 96k on validation and frozen test under k5/k20/k50. Cross-task-safe ranking summaries report point estimates, equalweight means, sample SD with $n=3$ and $ddof=1$, and per-seed directions. They describe training-run variability, not confidence intervals, significance, equivalence, or a multi-seed trajectory.

Candidate-generation seed42 is recorded separately from the three training seeds.

VI. RESULTS

We first compare the capabilities measured under Y and N, then examine exposure response, specialist-multitask relationships, candidate protocols, and the N-SASRec comparison. MovieLens supplies the full seed42 trajectories and independent seed43/44 replications at 96k. Validation guided decisions; frozen-test results are report-only held-out evidence. Amazon provides an earlier-point ranking-side check.

TABLE IV MovieLens seed42 supervision profiles. Panel A reports Y-native preference metrics across validation exposure and the held-out Y96 endpoint. Panel B reports PopMatch-k5 Y-as-ranker validation exposure, the Y96/N96 validation contrast, and frozen-test endpoints. AUC/F1/accuracy and HR@1/NDCG@5/MRR belong to different interfaces and are not compared numerically across panels.

Panel	Run	Exposure	Split	Interface	Metric 1	Value 1	Metric 2	Value 2	Metric 3	Value 3
A	Y24	24000	validation	Y-native preference	AUC	0.77613	F1	0.77917	Accuracy	0.71909
A	Y48	48000	validation	Y-native preference	AUC	0.78161	F1	0.78484	Accuracy	0.72304
A	Y96	96000	validation	Y-native preference	AUC	0.78435	F1	0.77832	Accuracy	0.72353
A	Y96	96000	test	Y-native preference	AUC	0.78535	F1	0.77802	Accuracy	0.72211
B	Y24	24000	validation	Y-as-ranker bridge	HR@1	0.21568	NDCG@5	0.60027	MRR	0.47041
B	Y48	48000	validation	Y-as-ranker bridge	HR@1	0.21656	NDCG@5	0.59946	MRR	0.46948
B	Y96	96000	validation	Y-as-ranker bridge	HR@1	0.22115	NDCG@5	0.60307	MRR	0.47418
B	N96	96000	validation	N-native ranking	HR@1	0.62379	NDCG@5	0.83029	MRR	0.77324
B	Y96	96000	test	Y-as-ranker bridge	HR@1	0.20652	NDCG@5	0.59216	MRR	0.46005
B	N96	96000	test	N-native ranking	HR@1	0.61004	NDCG@5	0.82190	MRR	0.76220

This contrast separates rating-derived preference discrimination from selecting the next observed event. Because target construction, task data, prompts, and scoring interfaces differ jointly, it compares complete supervision formulations rather than isolating one semantic cause.

B. Task-Specific Response to Training Exposure

Y-native validation AUC rises from 0.7761 at 24k to 0.7844 at 96k. F1 and the Y-as-ranker metrics do not improve

A. Different Supervision and Observed Recommendation Capabilities

Y and N show distinct measured capability profiles. Y96 reaches 0.7844 validation AUC on native MovieLens preference prediction. Through the P(Yes) bridge, its PopMatch-k5 validation HR@1 is 0.2211, compared with 0.6238 for N96 at the same nominal task-sample exposure. Native preference metrics, the Y-as-ranker trajectory, and validation/frozen-test Y96-N96 endpoints appear in Table IV.

uniformly, so the measured Y response is limited or uneven rather than demonstrably saturated.

N-native PopMatch-k5 validation HR@1 rises from 0.5774 at N24 to 0.6516 at N200, an absolute gain of 0.0742; the test trajectory has the same direction. Figure 1 shows the validation trend, while Table V retains the exact HR@1 values for both splits.

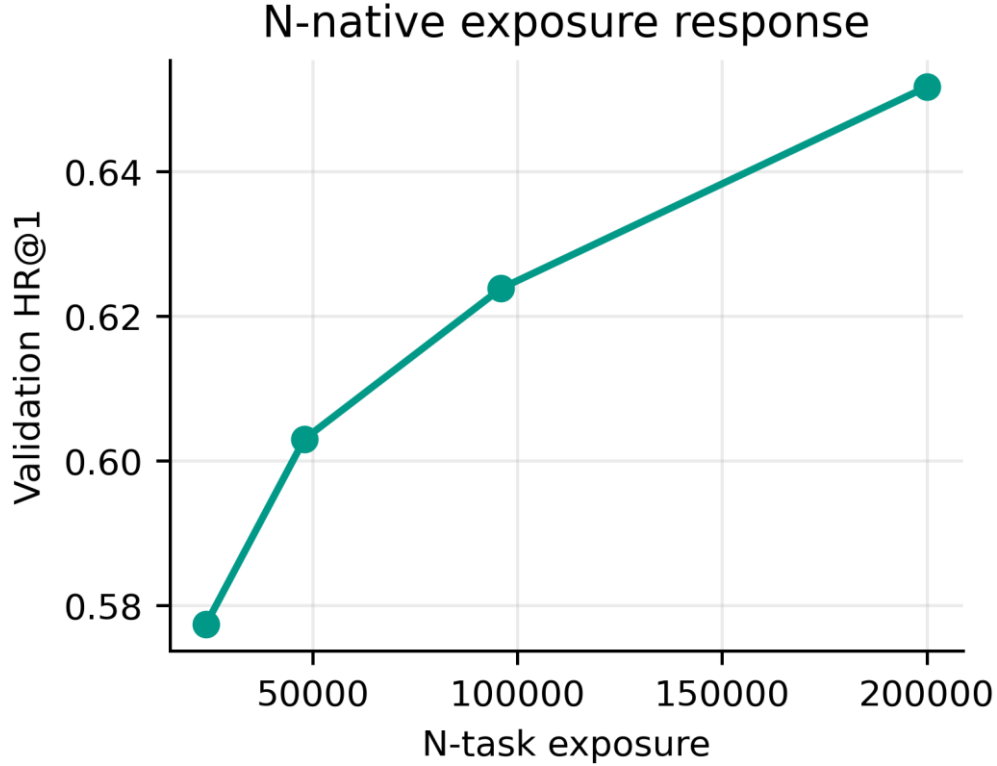


Fig. 1. MovieLens seed42 PopMatch-k5 N ranking performance across the evaluated task-exposure points.

TABLE V MovieLens seed42 N/SASRec ranking on validation and test sets at approximately matched task-sample exposure (HR@1).

N run	N exposure	SASRec exposure	Split	N HR@1	SASRec HR@1	Delta HR@1
N24	24000	24064	validation	0.57744	0.27313	0.30432
N24	24000	24064	test	0.56123	0.29040	0.27084
N48	48000	48128	validation	0.60300	0.29304	0.30996
N48	48000	48128	test	0.58696	0.30220	0.28476
N96	96000	96256	validation	0.62379	0.32811	0.29568
N96	96000	96256	test	0.61004	0.33093	0.27912
N200	200000	200000	validation	0.65163	0.47489	0.17674
N200	200000	200000	test	0.62819	0.45110	0.17709

Y and N therefore have different observed exposure-response profiles. This comparison concerns within-interface trajectories, not the numerical magnitudes of AUC and HR@1, and motivates matching corresponding task exposure in the specialist-multitask analysis.

C. Specialist and Multitask Models at Matched Per-Task Exposure

At 96k expected exposure per task, M1 retains comparable measured Y-side preference capability across three training

seeds. M1-Y validation AUC and F1 are slightly higher in each seed, whereas accuracy is lower by 0.00089 for seed44. Table VI reports the seed42 paired user bootstrap for this binary Y-side comparison: only the F1 interval is positive, while AUC and accuracy intervals cross zero. This supports capability preservation, not equivalence or positive transfer across training seeds.

TABLE VI MovieLens seed42 validation paired binary differences and user-bootstrap 95% intervals for Y96 and M1-Y96. Delta is M1-Y96 minus Y96. This table supports only the Y-side preference comparison.

Comparison	Metric	Delta	CI low	CI high
Y96 vs M1-96-Y	AUC	0.00248	-0.00167	0.00674
Y96 vs M1-96-Y	F1	0.00553	0.00096	0.01025
Y96 vs M1-96-Y	Accuracy	0.00460	-0.00089	0.01035

On the common cross-task-safe validation subset in Table VII, the seed42 standard-k5 N-specialist-M1 gap narrows from

48k to 96k per task across all three metrics. The HR@1 gap falls from +0.01072 to +0.00132; NDCG@5 and MRR show

the same direction. Independent seeds 43 and 44 reproduce a small but consistently positive N advantage at 96k: Table VIII reports a three-seed validation HR@1 mean delta of +0.00708

with sample SD 0.00934 and 3/3 positive directions. The full exposure trajectory remains seed42-only.

TABLE VII Seed42 common cross-task-safe PopMatch-k5 comparison. Delta is N minus M1; change is delta96 minus delta48 within each split and metric.

Split	Exp. (k)	n	Metric	N	M1-N	Delta	Delta change
validation	48	5318	HR@1	0.60323	0.59252	0.01072	
validation	96	5318	HR@1	0.62223	0.62091	0.00132	-0.00940
validation	48	5318	NDCG@5	0.82000	0.81451	0.00549	
validation	96	5318	NDCG@5	0.82961	0.82784	0.00177	-0.00372
validation	48	5318	MRR	0.75953	0.75225	0.00728	
validation	96	5318	MRR	0.77233	0.77004	0.00230	-0.00498
test	48	5535	HR@1	0.58500	0.57561	0.00939	
test	96	5535	HR@1	0.60741	0.59368	0.01373	0.00434
test	48	5535	NDCG@5	0.80952	0.80499	0.00453	
test	96	5535	NDCG@5	0.82044	0.81439	0.00605	0.00151
test	48	5535	MRR	0.74575	0.73974	0.00601	
test	96	5535	MRR	0.76028	0.75221	0.00807	0.00206

Frozen tests retain an N advantage in every seed at 96k. Table VIII reports a three-seed HR@1 mean delta of +0.01072 with sample SD 0.00271 and 3/3 positive directions. On the common seed42 test subset in Table VII, however, the HR@1 gap changes from +0.00939 at 48k to +0.01373 at 96k, and NDCG@5 and MRR likewise do not narrow. The validation-side exposure trajectory is therefore not reproduced on test.

D. Robustness under Alternative Candidate Protocols

At 96k, k20 produces the largest N-specialist-M1 gap among the three evaluated protocols. The cross-task-safe three-seed summary in Table VIII gives HR@1 mean deltas of +0.08042 on validation and +0.07678 on frozen test. Across k5, k20, and k50, all 54 seed-level deltas are positive and all 18 protocol-split-metric summaries have 3/3 positive seed directions.

TABLE VIII Cross-task-safe 96k N96-M1-N ranking deltas for training seeds 42/43/44. Mean and sample SD use n=3 and ddof=1; signs follow seed order. These are descriptive summaries. Protocols are separate and non-nested.

Split	Protocol	Metric	Delta	Mean	SD	Signs 42/43/44
validation	k5	HR@1	N96 - M1-N	0.00708	0.00934	+/+/+
validation	k5	NDCG@5	N96 - M1-N	0.00345	0.00288	+/+/+
validation	k5	MRR	N96 - M1-N	0.00461	0.00396	+/+/+
test	k5	HR@1	N96 - M1-N	0.01072	0.00271	+/+/+
test	k5	NDCG@5	N96 - M1-N	0.00528	0.00066	+/+/+
test	k5	MRR	N96 - M1-N	0.00700	0.00093	+/+/+
validation	k20	HR@1	N96 - M1-N	0.08042	0.02971	+/+/+
validation	k20	NDCG@5	N96 - M1-N	0.11826	0.01135	+/+/+
validation	k20	MRR	N96 - M1-N	0.08724	0.01896	+/+/+
test	k20	HR@1	N96 - M1-N	0.07678	0.03492	+/+/+
test	k20	NDCG@5	N96 - M1-N	0.11573	0.01356	+/+/+
test	k20	MRR	N96 - M1-N	0.08441	0.02143	+/+/+
validation	k50	HR@1	N96 - M1-N	0.00990	0.00696	+/+/+
validation	k50	NDCG@5	N96 - M1-N	0.01457	0.01310	+/+/+
validation	k50	MRR	N96 - M1-N	0.01830	0.00731	+/+/+
test	k50	HR@1	N96 - M1-N	0.01048	0.00444	+/+/+
test	k50	NDCG@5	N96 - M1-N	0.01418	0.01223	+/+/+
test	k50	MRR	N96 - M1-N	0.01806	0.00597	+/+/+

k50 also favors N across seeds, with three-seed HR@1 mean deltas of +0.00990 on validation and +0.01048 on test, but its relation to k5 is metric dependent. For seed43 test, k50 is smaller than k5 for HR@1 (+0.00560 versus +0.00849) and NDCG@5 (+0.00450 versus +0.00489), while its MRR delta is larger. k5 is popularity matched, whereas k20/k50 are random and all three protocols are separately constructed and non-nested. Candidate count, composition, sampling, and difficulty therefore vary jointly; the evidence establishes

protocol dependence, not a causal or monotonic candidate-size effect.

E. Exposure-Aware Comparison with SASRec

N-trained LlamaRec leads SASRec at all four approximately matched seed42 downstream task-sample exposure points. On validation, the HR@1 gap narrows from +0.2957 at 96k to +0.1767 at 200k. Table V retains exact exposures and validation/frozen-test values; Figure 2 shows the validation trajectories.

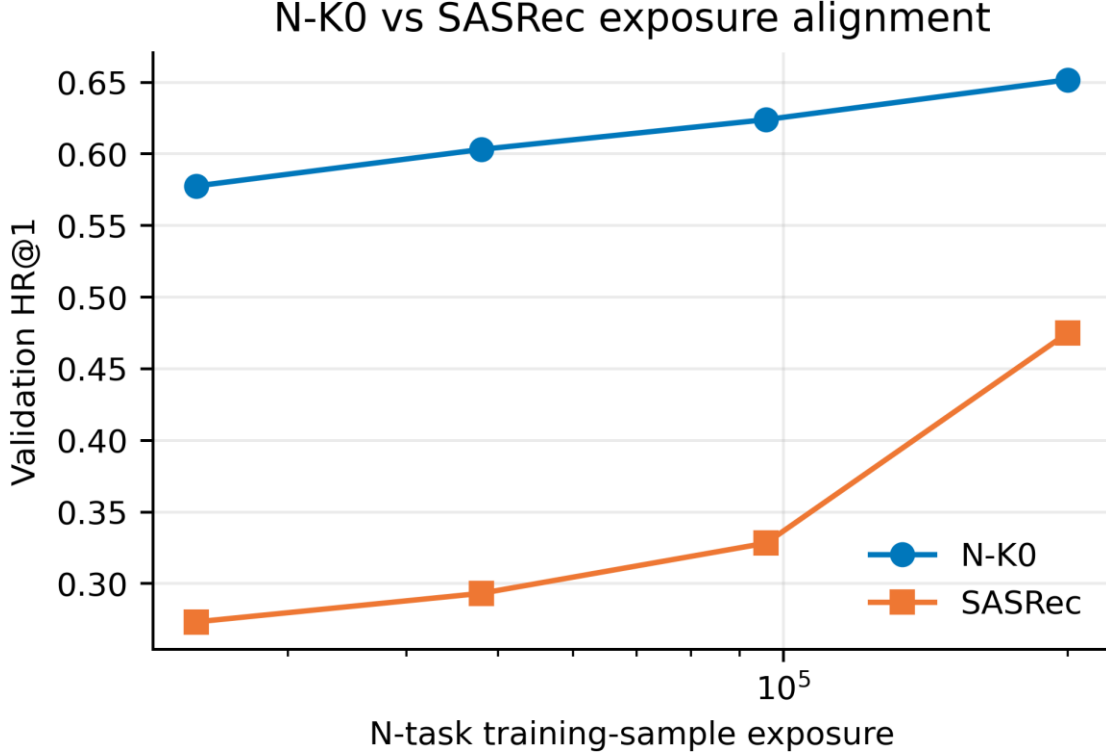


Fig. 2. MovieLens seed42 PopMatch-k5 validation HR@1 at approximately matched task-sample exposure. The N-K0 legend denotes the N24/N48/N96/N200 series.

SASRec nevertheless improves strongly with additional supervision. The comparison therefore pairs N’s lead with the baseline’s continuing response and aligns only downstream task-sample exposure, not pretraining, tokens, updates, FLOPs, time, or end-to-end resource efficiency.

F. Cross-Dataset Directional Validation

At the evaluated earlier seed42 operating points on Amazon Musical Instruments, N reaches 0.4669 HR@1 on 57,439 PopMatch-k5 test examples, compared with 0.3573 for Base, 0.2298 for Y-as-ranker, and 0.1757 for SASRec. Table IX reports these four cross-task-safe or task-specific model results with HR@1, NDCG@5, and MRR.

TABLE IX Amazon Musical Instruments seed42 full-test PopMatch-k5 ranking for the evaluated earlier operating points.

Run	Samples	HR@1	NDCG@5	MRR
base	57439	0.35730	0.68295	0.57896
y	57439	0.22979	0.60997	0.48303
n	57439	0.46688	0.74201	0.65698
sasrec	57439	0.17566	0.56851	0.42952

Amazon therefore provides ranking-side directional support at available earlier points, not a second exposure curve, three-seed 96k replication, hard-candidate study, or native-preference validation.

VII. DISCUSSION

The findings tie three parts of a recommendation comparison to one another: supervision formulation defines the prediction question, task exposure locates the adaptation point, and candidate protocol defines the measured ranking conditions. Their implications differ from the numerical results and are organized below by the study’s contribution hierarchy.

A. Supervision Formulation and Capability Interpretation

Recommendation capability is meaningful only relative to a prediction question. Y estimates rating-derived preference, whereas N selects the next observed interaction. Native preference discrimination and bridge ranking therefore assess different behavior: a model can be useful for preference while being less suited to next-event selection.

A benchmark’s target, eligible events, prompt, and scoring rule jointly determine the capability reflected by its score. Reporting native and bridge results exposes distinctions that one ranking number can hide and keeps the Y/N contrast attached to complete formulations rather than a single semantic cause.

B. Exposure and Specialist-Shared Modeling

Exposure locates a comparison along the observed adaptation trajectory. Seed42 shows continued N improvement but limited or uneven Y gains, so an early checkpoint need not describe the later model relationship. Treating task exposure as an experimental variable prevents an operating point from being mistaken for a stable property of the task.

Task-specific accounting distinguishes M1’s total exposure from the supervision allocated to each interface and makes SASRec comparisons interpretable across batch sizes. This axis is cumulative downstream consumption, including repetitions, rather than unique data or computation. Stating it beside performance makes the budget condition of each advantage visible.

The standard-k5 validation trajectory and independent 96k replications together show that specialist-shared relations are conditional on adaptation point. M1 retains comparable Y-side capability at 96k, while frozen tests still favor N. A shared adapter should therefore be assessed by capability retained at a stated exposure and protocol, not by an unconditional success/failure label. The full trajectory remains seed42-only.

C. Candidate Protocol and Comparison Scope

At 96k, k20 exposes a larger N advantage than standard k5, whereas k50 is smaller than k20 and not uniformly larger than k5. A standard-k5 result alone therefore cannot characterize specialist performance retained under other candidate conditions. The relation is protocol-conditioned, not monotonic in candidate count.

Fixed candidates within a protocol support paired model comparison, while cross-protocol differences describe sensitivity to evaluated conditions. Because the sets are independently constructed and non-nested, count, composition, sampling, and difficulty change jointly. This separates within-protocol evidence from unsupported candidate-count causality.

D. Exposure-Aligned Cross-Model Baseline Comparison

Across four approximately matched seed42 points, N leads the evaluated SASRec baseline, while SASRec improves substantially and narrows the largest-exposure gap. The former describes performance under limited downstream task supervision; the latter describes the baseline’s response to receiving more supervision. Both are needed to interpret the cross-model comparison.

Task-sample exposure is a useful but partial cross-model axis. It describes adaptation from the LLM’s pretrained starting point and does not match pretraining, tokens, updates, FLOPs, time, serving, or deployment cost. The evidence supports an exposure-aligned performance comparison, not general resource efficiency.

MovieLens supplies the detailed within-domain exposure and protocol analysis, whereas Amazon checks ranking directions only at available earlier seed42 points. The two domains therefore provide complementary detailed and directional evidence; untested cross-domain trajectories remain outside the present claims.

Recommendation capability and relative performance should thus be interpreted within the evaluated formulation,

cumulative per-task exposure, and candidate protocol. The cross-model trajectory adds that an observed N advantage can coexist with continued improvement of the sequential baseline.

VIII. LIMITATIONS

Only the 96k operating point has independent training seeds 42/43/44; full MovieLens trajectories remain seed42-only. Cross-task-safe ranking means and sample SD describe $n=3$ training-run variability and have no bootstrap intervals, whereas the retained seed42 paired bootstrap describes evaluation-sample uncertainty only for the fixed Y-side models. Neither establishes cross-seed significance. Validation guided selection and frozen test provides held-out reporting. No equivalence margin or allcomparison multiplicity adjustment was used. Behavior beyond Y96/N200 is unknown, and exposure changes coverage and optimization history together. M1 per-task counts are schedule expectations conditional on correct resume skipping, without a retained per-example trace.

Shared-model ranking comparisons use high-coverage cross-task-safe subsets rather than the entire N holdouts because Y and N use different task-specific temporal splits. This evaluation-side restriction ensures that an N target is not exposed through the shared adapter’s Y-training stream; M1 was not trained under a joint temporal cutoff. Retained coverage ranges from 93.71% to 98.68%, so the conclusions apply to these retained populations. PopMatch-k5 and random k20/k50 are non-nested: count, composition, sampling, and difficulty vary jointly. PopMatch uses full-corpus popularity retrospectively, and sampled ranking is not exhaustive full-catalog evaluation.

Amazon support is limited to seed42 ranking directions at earlier test operating points. It does not replicate exposure trajectories, the three-seed 96k point, hard-candidate robustness, or native preference validation. Detailed exposure and protocol conclusions therefore remain specific to MovieLens.

Task-sample matching does not remove resource differences. The LLM starts pretrained; SASRec learns sequence representations from recommendation data. Exposure matches neither pretraining nor tokens, updates, FLOPs, wall-clock time, serving latency, or deployment cost. Claims concern ranking under approximately matched downstream task-sample exposure in the evaluated implementations and regimes.

IX. CONCLUSION

This study examined LLM recommendation conclusions across complete supervision formulations, cumulative downstream task-sample exposure, and candidate protocols. Within one adaptation setting, rating-derived preference (Y), next-interaction prediction (N), and a shared adapter (M1) represent different prediction capabilities rather than a causally isolated semantic contrast.

On the common cross-task-safe subset, the seed42 specialist-M1 validation gap narrows from 48k to 96k across all three ranking metrics, while the test subset does not reproduce that narrowing. Independent seeds replicate only the 96k operating point, where ranking consistently favors N and

M1 preserves comparable Y-side capability. Thus, specialist-shared performance is exposure and protocol conditioned rather than universally equivalent. Across four approximately matched downstream exposure points, N-trained LlamaRec leads SASRec, but SASRec continues improving and narrows the later gap; this is a task-sample exposure comparison, not a resource-efficiency claim.

Claims about adapted language models for recommendation should therefore report the prediction formulation, cumulative downstream task-sample exposure, and candidate evaluation protocol together. These conditions define which capability and relative model performance the evidence supports.

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